Probability

→ likelihood of an event to occur.

Ex:- Tossing a coin.

$P(H) = \frac{1}{2} = 50\%$

* Mutual Exclusive event :-

Two events are mutual exclusive if they cannot occur at the same time.

Ex:- Tossing a coin

$$\boxed{P(H \text{ or } T) = P(H) + P(T)}$$ → Additive rule for mutually exclusive

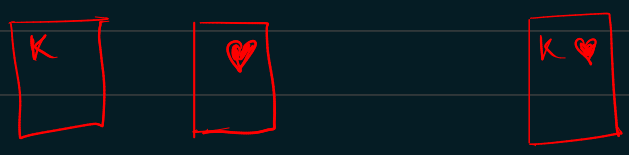
Ex:- Rolling a dice. $\{1, 2, 3, 4, 5, 6\}$

$P(1 \text{ or } 5) = P(1) + P(5)$

$= 1/6 + 2/6 = 1/3$

* Non-mutual exculsive events :-

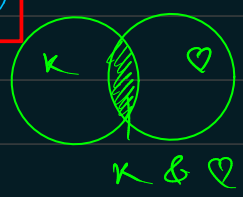
Ex:- Taking a card from the deck.



Additive rule for non-mutual exclusive events.

$$\boxed{P(K \text{ or } ♡) = P(K) + P(♡) - P(K \text{ \& } ♡)}$$

$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = 16/52$



Multiplication Rule

Independent event :-

2 events are independent if they dont effect each other.

Ex :- Tossing a coin.

$P(H) = 1/2 \rightarrow$ later $\rightarrow P(T) = 1/2$

Dependent events :-

2 events are dependent if they effect each other.

Ex :- Take a king from deck & then queen card from deck

52

$P(K) = 4/52 \longrightarrow P(Q) = \frac{4}{52}$

① for independent event

$$\boxed{P(H \text{ and } T) = P(H) \cdot P(T)}$$

$$= 1/2 \cdot 1/2 = \frac{1}{4}$$

② for dependent event

$$\boxed{P(K \,\&\, Q) = P(K) \cdot P(Q/K)} \rightarrow \text{Conditional probability}$$

$$= \frac{4}{52} \cdot \frac{4}{51}$$